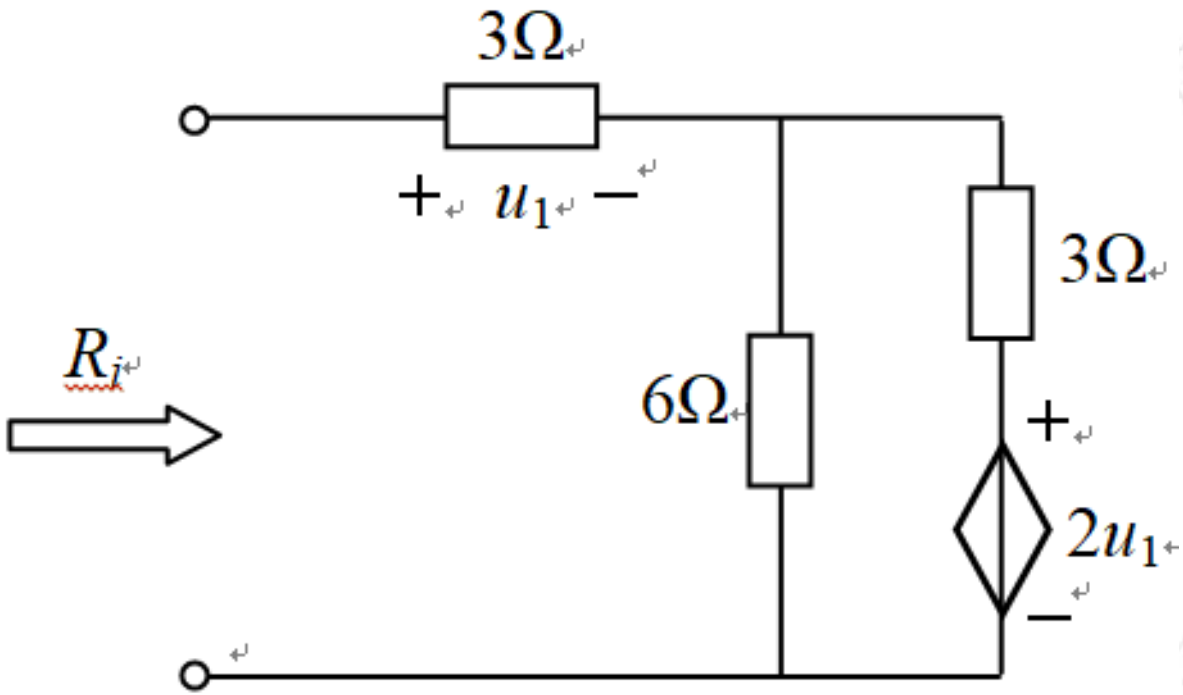
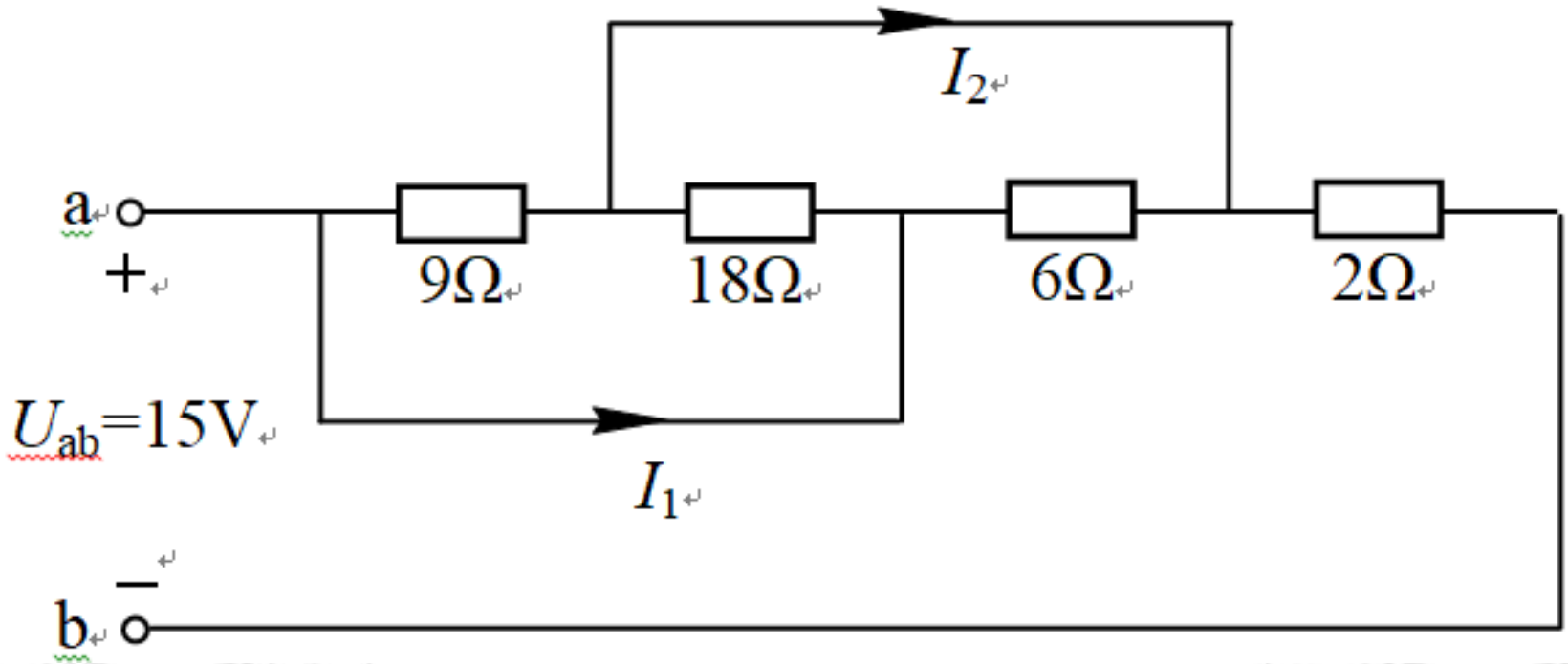
试求输入电阻R_i。



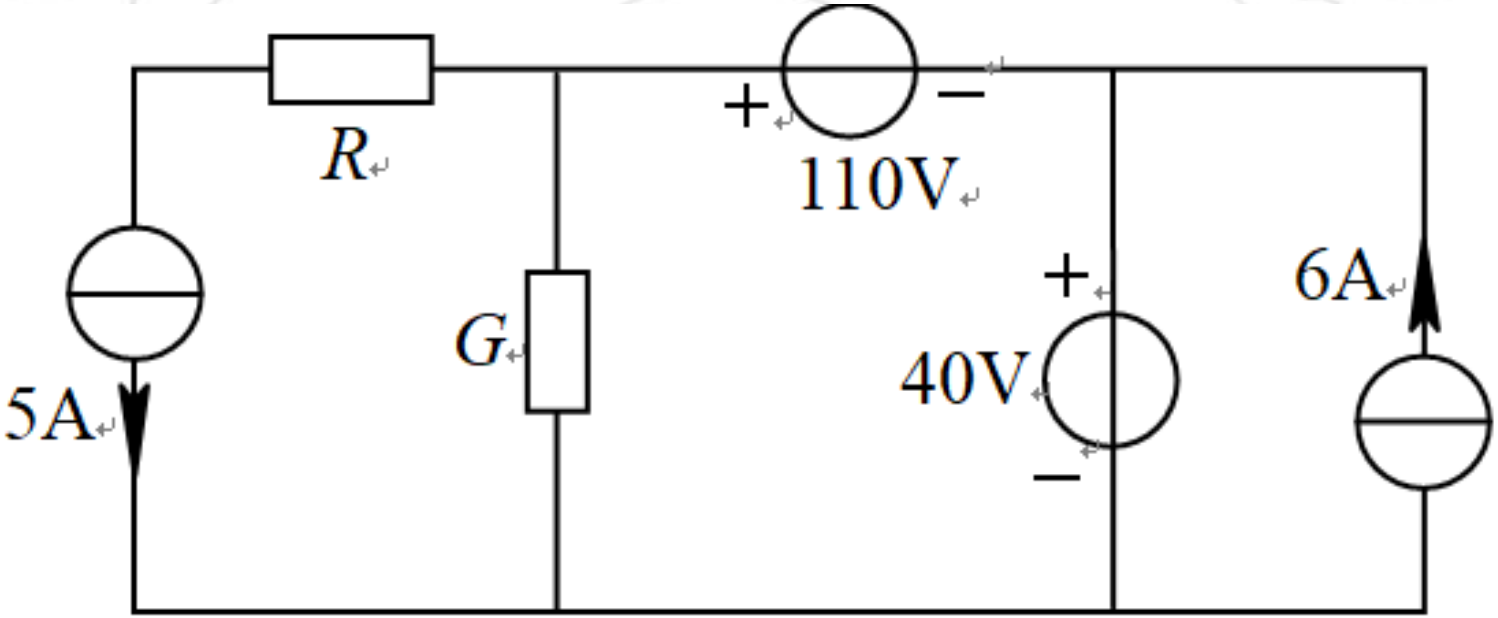
$$R_i = 9\Omega$$

试求等效电阻R_{ab}及电流I_1和I_2。



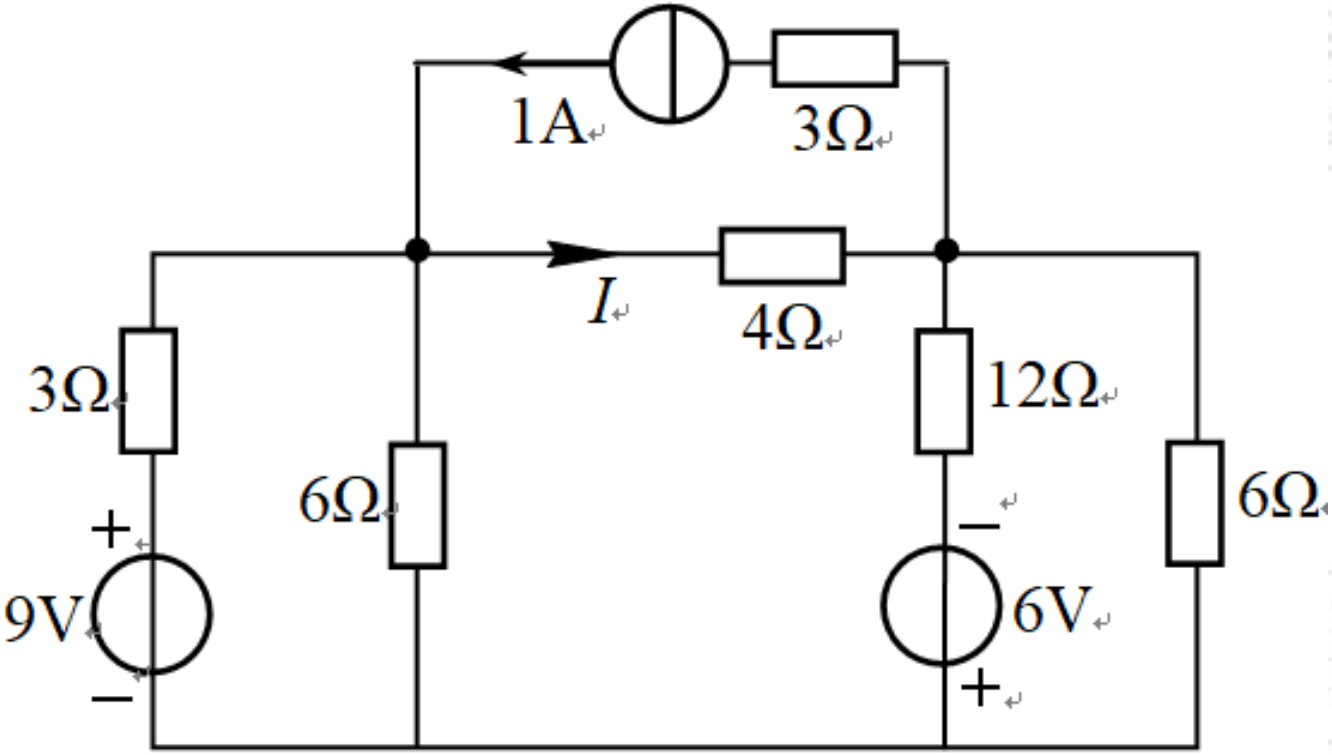
$R_{ab} = 5\Omega$　　$I_1 = 2\text{A}$　　$I_2 = 1.5\text{A}$

已知5A电流源提供100W功率，40V电压源提供500W功率，求*R*和*G*。



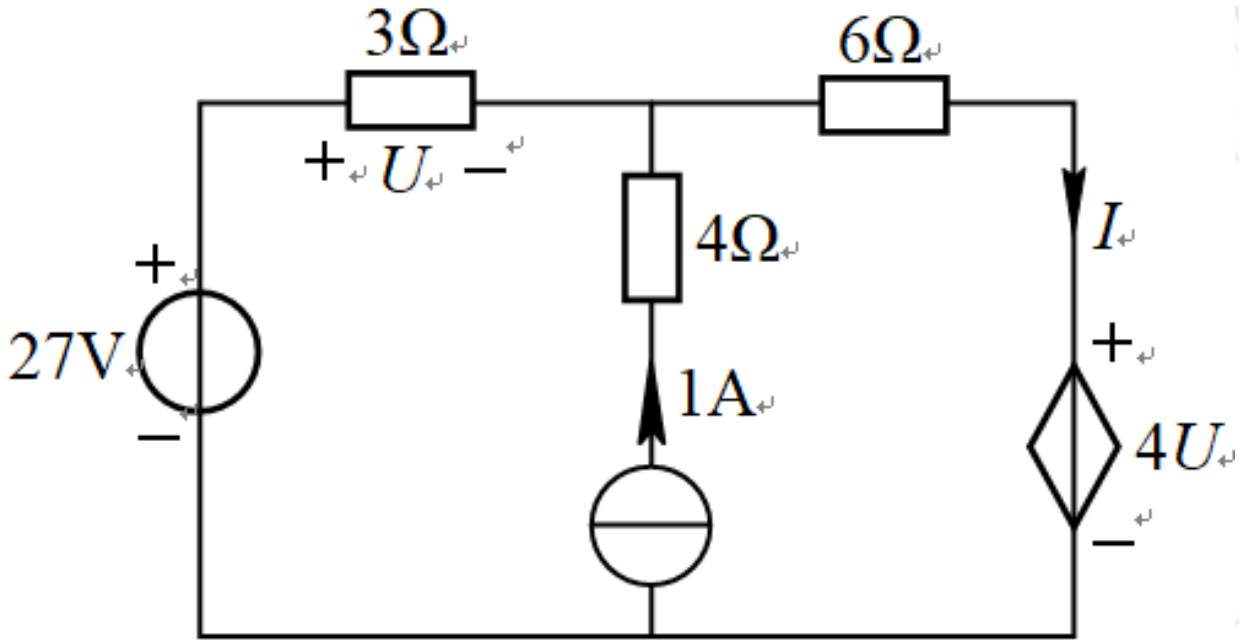
$R = 34\Omega$　　$G = 0.09S$

试用电源等效变换的方法求电流I。



$$I = 1.4\text{A}$$

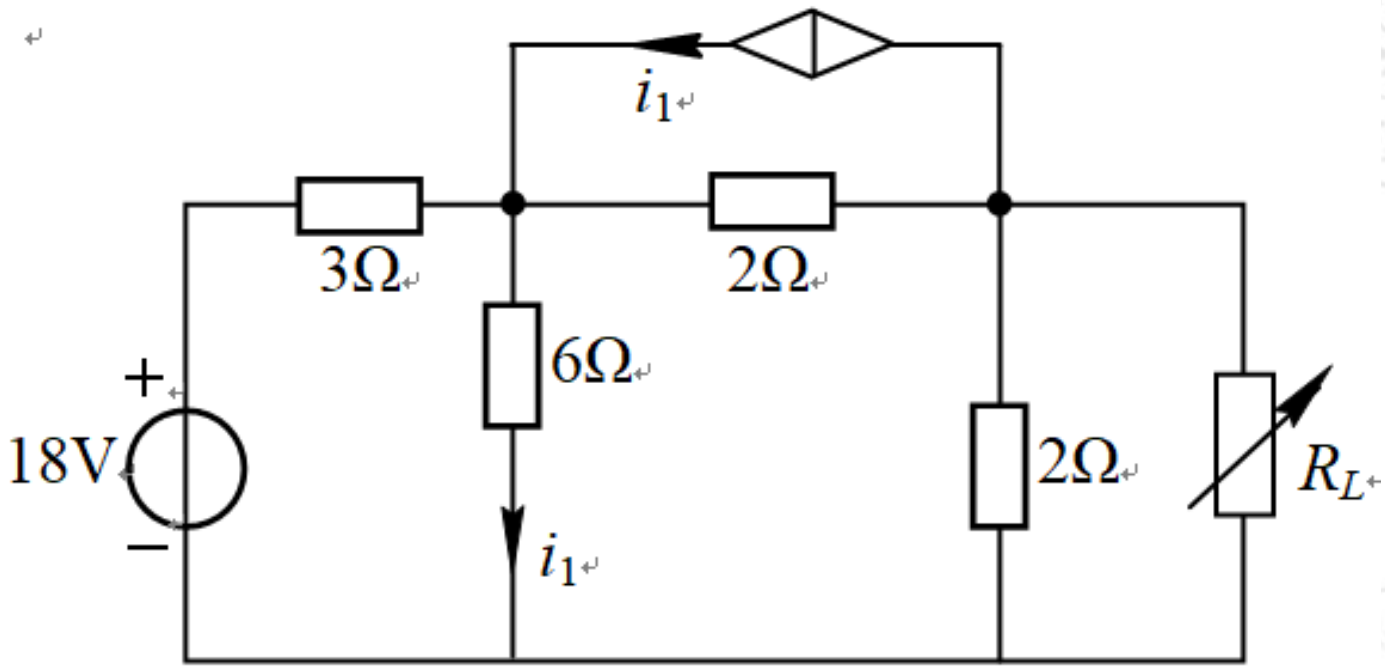
试用叠加定理求*U*和*I*的值。



$$U = \frac{27}{7} - \frac{6}{7} = 3\text{V}$$

$$I = \frac{9}{7} + \frac{5}{7} = 2\text{A}$$

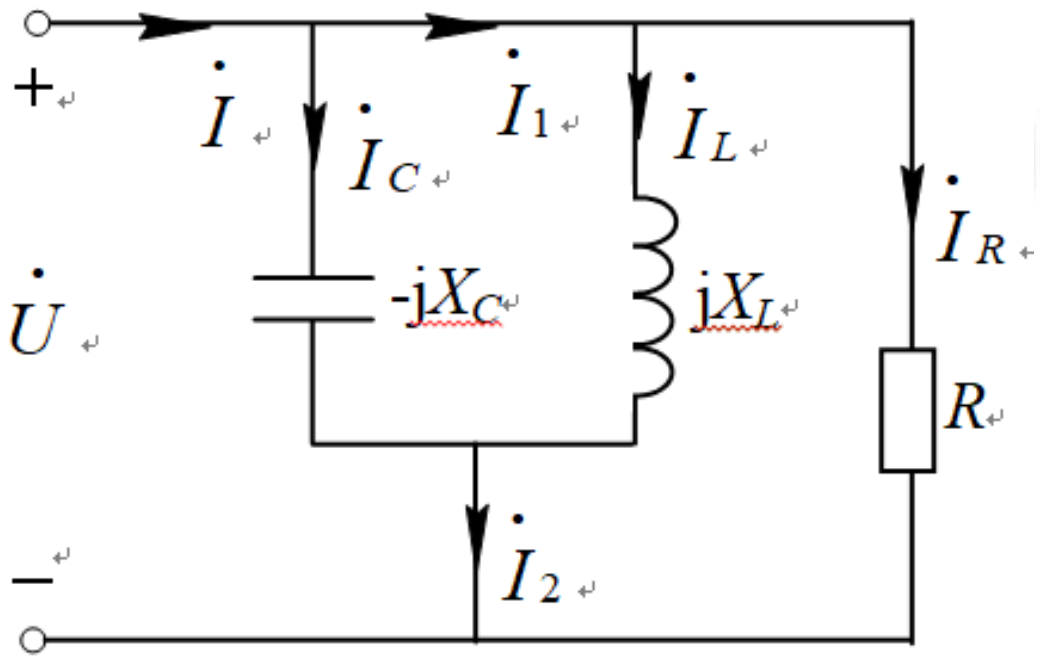
$R_L=?$时可获最大功率，并求此最大功率值P_{max}。



$$U_{OC}^{*}=8\text{V},\ R_{eq}^{*}=\frac{10}{3}\Omega$$

$$\frac{5}{4}\Omega,\ 1.8\text{W}$$

正弦稳态电路如图所示，电流有效值 $I = I_C = I_1 = 10\text{A}$，电路消耗平均功率 $P = 1500\text{W}$。试以 $\dot{U}$ 为参考相量画电路相量图，并求电阻 R、感抗 X_L、容抗 X_C 及电流有效值 I_2。



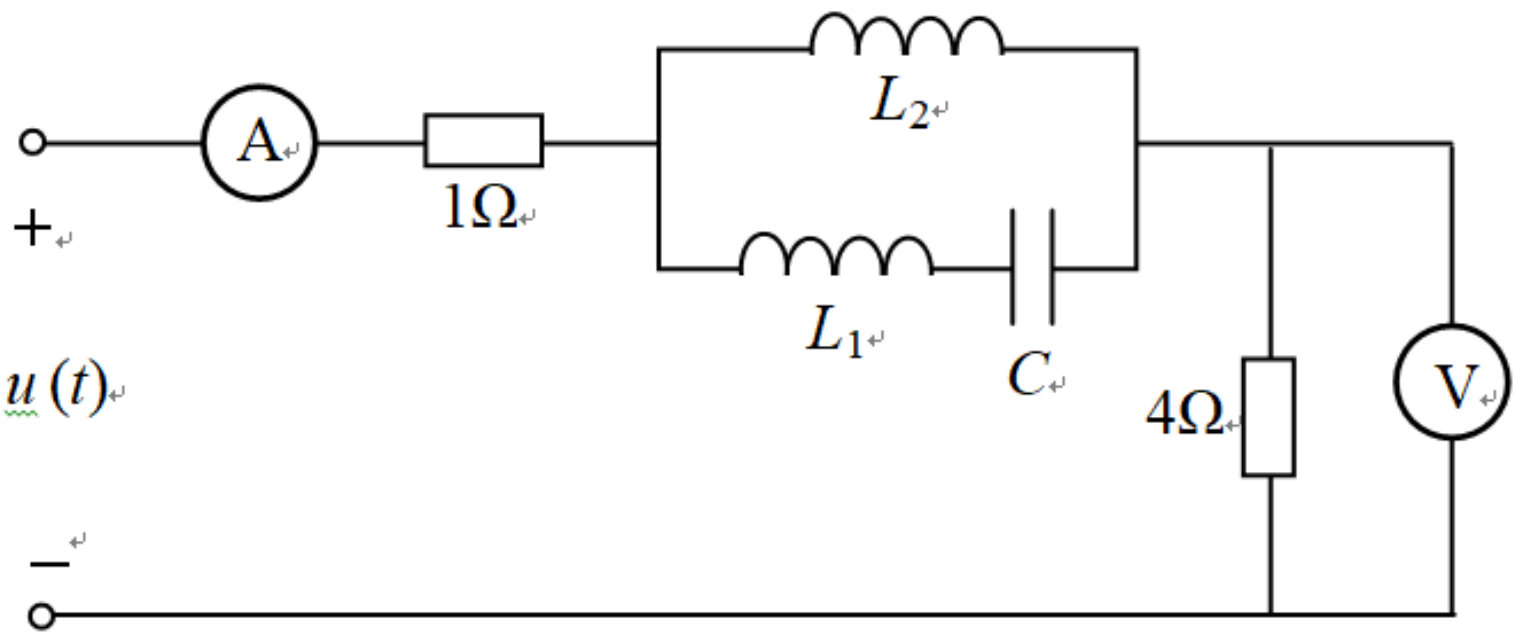
$$R = 20\Omega$$

$$X_C = 10\sqrt{3}\Omega$$

$$X_L = 20\sqrt{3}\Omega$$

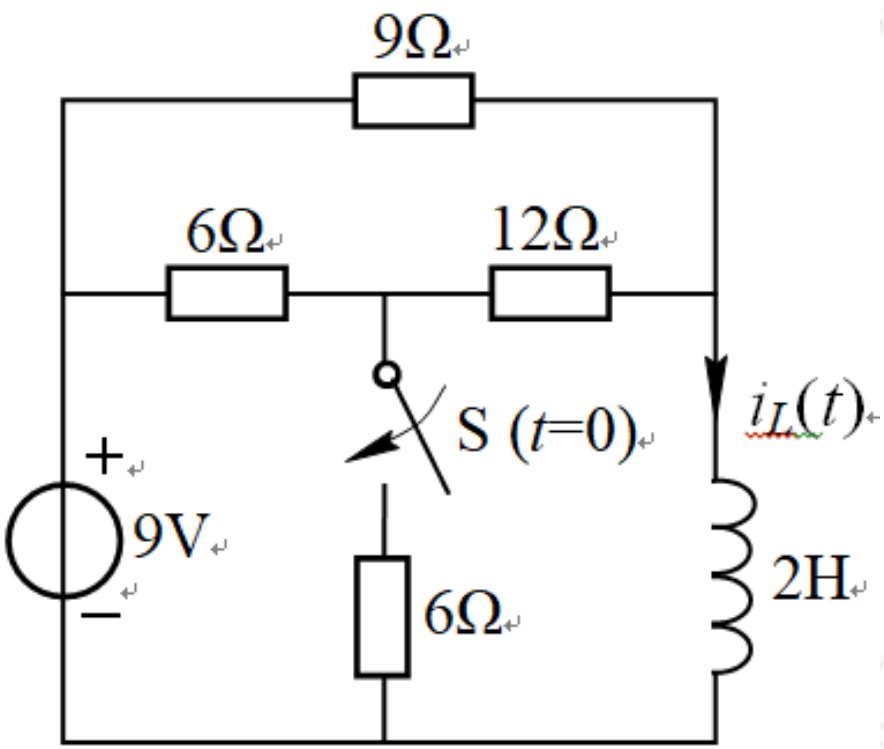
$$I_2 = 5\text{A}$$

非正弦周期电流电路如图所示，电压表、电流表的读数均为有效值，已知$u(t)=20+20\sqrt{2}\cos\omega t+15\sqrt{2}\cos(2\omega t+90^\circ)\text{V}$，$\omega L_1=10\Omega$，$\omega L_2=30\Omega$，$\dfrac{1}{\omega C}=40\Omega$。试求电压表和电流表的读数。



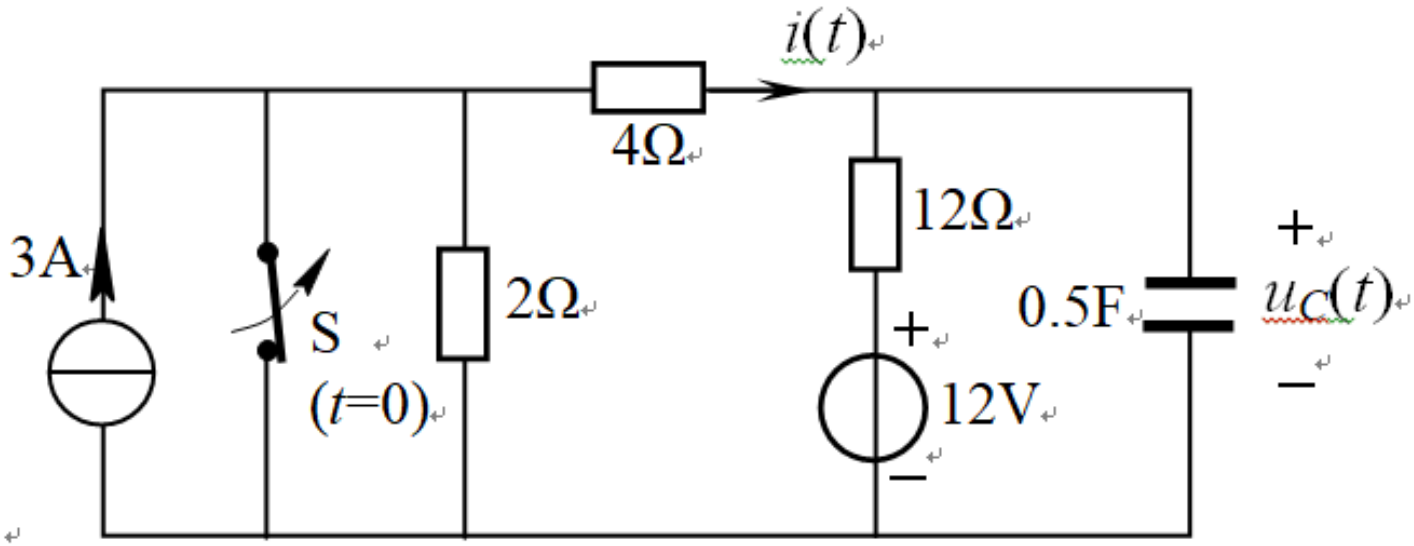
$I=5\text{A}$　　$U=20\text{V}$

一阶电路如图所示，$t<0$ 时原电路已稳定，$t=0$ 时合上开关 S，试求 $t \geq 0_+$ 时的 $i_L(t)$，并画出其变化曲线。



$$i_L(t) = 1.3 + 0.2e^{-\frac{45}{16}t}\ \mathrm{A}\ (t \geq 0_+)$$

一阶电路如图所示，$t<0$ 时原电路已稳定，$t=0$ 时打开开关 S。试求 $t \geq 0_+$ 时的 $u_C(t)$，$i(t)$，并定性画出其变化曲线。



$$u_C(t) = 8 - 5e^{-\frac{t}{2}}\text{V}\ (t \geq 0_+)$$

$$i = -\frac{1}{3} + \frac{5}{6}e^{-\frac{t}{2}}\text{A}\ (t \geq 0_+)$$